ADVANCED WIRELESS COMMUNICATIONS SUMMARY (NCC325)

MODULE 1

INTRODUCTION TO ADVANCED WIRELESS COMMUNICATIONS

1.1 Definition of Wireless Communication

Wireless communication is the transmission of information (voice, data, video, or signals) **without physical connecting cables**, using **electromagnetic waves** such as radio waves, microwaves, and infrared waves.

In wireless systems, information is sent from a **transmitter** to a **receiver** through a wireless medium called the **radio channel**.

1.2 Evolution of Wireless Communication Systems

(a) First Generation (1G)

- Analog communication system
- Used mainly for voice
- Poor quality and security
- Example: AMPS

(b) Second Generation (2G)

- Digital communication
- Improved voice quality
- Introduced SMS
- Examples: GSM, CDMA

(c) Third Generation (3G)

- Supports voice and data
- Higher data rates
- Mobile internet introduced
- Example: UMTS

(d) Fourth Generation (4G)

- High-speed data transmission
- Supports multimedia services
- Uses IP-based communication
- Example: LTE

(e) Fifth Generation (5G)

- Very high data rates
- Low latency

- Supports IoT and smart devices
- Used in smart cities and autonomous systems

1.3 Advantages of Wireless Communication

1. Mobility and flexibility
2. Easy installation
3. Cost-effective for large coverage
4. Supports real-time communication
5. Enables global connectivity

1.4 Limitations of Wireless Communication

1. Susceptible to interference and noise
2. Limited bandwidth
3. Security challenges
4. Signal attenuation and fading
5. Power constraints in mobile devices

1.5 Applications of Wireless Communication

- Mobile telephony
- Internet access (Wi-Fi, mobile data)
- Broadcasting (radio and television)
- Satellite communication
- Emergency services
- Smart homes and IoT

1.6 Basic Components of a Wireless Communication System

(a) Transmitter

- Converts information into a suitable signal
- Performs modulation

(b) Channel

- Medium through which the signal travels
- Affected by noise and interference

(c) Receiver

- Receives and demodulates the signal
- Recovers original information

1.7 Challenges in Wireless Communication

- Multipath propagation

- Doppler effect due to movement
- Interference from other users
- Limited power and spectrum resources

1.8 Wireless Network Architecture

- Base Station (BS)
- Mobile Station (MS)
- Core Network
- Backhaul network

1.9 Spectrum Allocation and Regulation

- Wireless communication uses **licensed and unlicensed frequency bands**
- Regulated by national and international bodies
- In Nigeria: **Nigerian Communications Commission (NCC)**
- Globally: **International Telecommunication Union (ITU)**

1.10 Advantages and Disadvantages of Wireless Networks vs Wired Networks

Advantages:

- Mobility and flexibility for users.
- Easy to expand network coverage without physical cabling.
- Quick installation in areas where cabling is impractical.
- Supports mobile devices like laptops, smartphones, and IoT devices.

Disadvantages:

- Slower speeds and higher latency than wired networks.
- Signal interference from other devices or obstacles.
- Less secure; vulnerable to eavesdropping and attacks.
- Limited range; coverage decreases with distance or walls.

1.11 Differentiate Wireless and Wired Networks

Feature	Wired Network	Wireless Network
Medium	Uses physical cables (Ethernet, fiber optics, coaxial)	Uses radio waves, infrared, or microwaves
Mobility	Devices are fixed; moving requires reconnection	High mobility; devices can move freely within coverage
Installation Cost	Higher cost due to cables and labor	Lower physical installation cost; may require access points

Feature	Wired Network	Wireless Network
Data Speed & Reliability	Typically faster and more stable	Generally slower; affected by interference and distance
Security	More secure due to physical access restriction	More vulnerable to eavesdropping and attacks
Maintenance	Requires cable management and repair	Requires monitoring of signals, frequencies, and interference

1.12 Compare 2G and 3G Mobile Technologies

Feature	2G	3G
Introduced	Early 1990s	Early 2000s
Data Speed	Up to 64 Kbps	Up to several Mbps
Service Type	Voice-centric, SMS support	Multimedia support: video calls, internet, email
Technology	GSM, CDMA	UMTS, CDMA2000
Network Capability	Basic text and voice	Enhanced mobile internet, mobile TV, video calls
Security	Basic encryption	Stronger encryption and authentication

MODULE 2

WIRELESS PROPAGATION AND CHANNEL CHARACTERISTICS

2.1 Introduction

Wireless propagation refers to **how radio signals travel from a transmitter to a receiver** through space. Unlike wired communication, wireless signals are affected by **distance, obstacles, environment, noise, and interference**.

The wireless channel is **unpredictable and time-varying**, making signal propagation a major challenge in wireless communication systems.

2.2 Free Space Propagation Model

Free space propagation assumes **no obstacles** between transmitter and receiver.

Characteristics:

- Signal strength decreases with distance
- Used for satellite and line-of-sight communication
- Ideal and theoretical model

Free Space Path Loss:

- Power received decreases as distance increases

- Higher frequency → higher path loss

2.3 Path Loss

Path loss is the **reduction in signal power** as it travels through space.

Causes of Path Loss:

1. Distance between transmitter and receiver
2. Absorption by objects (buildings, trees)
3. Reflection and diffraction

Types of Path Loss Models:

- Free space model
- Urban model
- Suburban model
- Indoor model

2.4 Shadowing

Shadowing is the **slow variation in received signal strength** caused by large obstacles like:

- Buildings
- Hills
- Trees

Characteristics:

- Also called **large-scale fading**
- Occurs over long distances
- Causes signal blockage

2.5 Multipath Propagation

Multipath propagation occurs when transmitted signals reach the receiver through **multiple paths** due to:

- Reflection
- Diffraction
- Scattering

Effects of Multipath:

- Signal distortion
- Inter-symbol interference (ISI)
- Fading

2.6 Fading

Fading is the **variation in signal strength** over time or space.

Types of Fading:

(a) Fast Fading

- Occurs due to multipath
- Rapid changes in signal strength
- Affects short distances

(b) Slow Fading

- Caused by shadowing
- Occurs over long distances

2.7 Doppler Effect

The Doppler effect is the **change in signal frequency** due to relative movement between the transmitter and receiver.

Effects:

- Causes frequency shift
- Leads to signal distortion
- Common in mobile users

2.8 Noise in Wireless Communication

Noise is any **unwanted disturbance** that affects signal quality.

Types of Noise:

1. Thermal noise
2. Interference noise
3. Atmospheric noise

2.9 Interference

Interference occurs when **multiple signals overlap** in frequency or time.

Sources of Interference:

- Co-channel interference
- Adjacent channel interference
- Other wireless devices

2.10 Wireless Channel Characteristics (Summary)

- Time-varying
- Frequency selective
- Random in nature
- Affected by environment

MODULE 3

DIGITAL MODULATION TECHNIQUES

3.1 Introduction

Digital modulation is the process of **converting digital data into a modulated carrier signal** suitable for transmission over a wireless channel.
It allows efficient use of bandwidth, better noise immunity, and reliable communication.

3.2 Need for Digital Modulation

Digital modulation is used in wireless communication because it:

1. Supports high data rates
2. Provides better noise resistance
3. Enables efficient spectrum utilization
4. Supports encryption and security
5. Works well with digital devices

3.3 Basic Digital Modulation Techniques

(a) Amplitude Shift Keying (ASK)

In ASK, the **amplitude of the carrier signal** is varied according to the digital data.

- Binary 1 → High amplitude
- Binary 0 → Low or zero amplitude

Advantages:

- Simple to implement

Disadvantages:

- Highly sensitive to noise

(b) Frequency Shift Keying (FSK)

In FSK, the **frequency of the carrier signal** is varied.

- Binary 1 → Higher frequency
- Binary 0 → Lower frequency

Advantages:

- Better noise immunity than ASK

Disadvantages:

- Requires more bandwidth

(c) Phase Shift Keying (PSK)

In PSK, the **phase of the carrier signal** is varied according to the data.

- Binary PSK (BPSK) uses two phase states

Advantages:

- Good noise performance

Disadvantages:

- More complex than ASK and FSK

3.4 Quadrature Phase Shift Keying (QPSK)

QPSK uses **four different phase shifts** to represent two bits per symbol.

Features:

- Higher data rate than BPSK
- Efficient bandwidth usage
- Widely used in cellular systems

3.5 Quadrature Amplitude Modulation (QAM)

QAM combines **amplitude and phase modulation**.

Features:

- Higher data transmission rate
- Examples: 16-QAM, 64-QAM
- Used in LTE, Wi-Fi

Limitation:

- More sensitive to noise

3.6 Bit Error Rate (BER)

BER is the **number of bit errors divided by total transmitted bits**.

Importance:

- Measures system performance
- Lower BER → better communication quality

3.7 Introduction to OFDM

Orthogonal Frequency Division Multiplexing (OFDM) divides data into **multiple subcarriers**.

Advantages:

- Resistant to multipath fading
- Efficient spectrum usage
- Used in Wi-Fi, LTE, and 5G

3.8 Comparison of Digital Modulation Techniques

Technique	Parameter Changed	Noise Immunity	Bandwidth Efficiency
ASK	Amplitude	Low	Low

Technique	Parameter Changed	Noise Immunity	Bandwidth Efficiency
FSK	Frequency	Medium	Low
PSK	Phase	High	Medium
QAM	Amplitude & Phase	Medium	High

MODULE 4

MULTIPLE ACCESS TECHNIQUES

4.1 Introduction

Multiple access techniques are methods used to **allow many users to share the same communication channel** without causing unacceptable interference.

In wireless systems, limited frequency spectrum makes multiple access techniques very important.

4.2 Need for Multiple Access Techniques

Multiple access is required to:

1. Allow many users to communicate simultaneously
2. Efficiently utilize available bandwidth
3. Reduce interference between users
4. Improve system capacity
5. Support cellular and wireless networks

4.3 Frequency Division Multiple Access (FDMA)

In FDMA, the total available bandwidth is divided into **several frequency bands**, and each user is assigned a different frequency.

Features:

- Users transmit continuously
- Each user has a unique frequency

Advantages:

- Simple to implement
- Low delay

Disadvantages:

- Inefficient bandwidth usage
- Guard bands reduce spectrum efficiency

4.4 Time Division Multiple Access (TDMA)

In TDMA, users share the same frequency but transmit in **different time slots**.

Features:

- Each user transmits in a short time period
- Requires synchronization

Advantages:

- Better bandwidth utilization than FDMA
- Flexible allocation

Disadvantages:

- Synchronization complexity
- Time slot delay

4.5 Code Division Multiple Access (CDMA)

In CDMA, all users transmit over the **same frequency band at the same time** but use **different spreading codes**.

Features:

- Uses spread spectrum technique
- High resistance to interference

Advantages:

- High capacity
- Improved security
- Better resistance to multipath fading

Disadvantages:

- Complex receiver design
- Near–far problem

4.6 Orthogonal Frequency Division Multiple Access (OFDMA)

OFDMA is an advanced version of FDMA where the channel is divided into **multiple orthogonal subcarriers**.

Features:

- Users are assigned subsets of subcarriers
- Used in LTE and WiMAX

Advantages:

- High spectral efficiency
- Robust against multipath fading

Disadvantages:

- High system complexity

- Sensitive to frequency offset

4.7 Comparison of Multiple Access Techniques

Technique	Division Method	Bandwidth Efficiency	Complexity
FDMA	Frequency	Low	Low
TDMA	Time	Medium	Medium
CDMA	Code	High	High
OFDMA	Frequency & Time	Very High	High

4.8 Application of Multiple Access Techniques

- FDMA: Early analog systems
- TDMA: GSM
- CDMA: 3G systems
- OFDMA: 4G LTE and WiMAX

MODULE 5

CELLULAR COMMUNICATION SYSTEMS

5.1 Introduction

A cellular communication system is a wireless communication system that **divides a large geographical area into smaller regions called cells**. Each cell is served by a **Base Station (BS)** to provide communication services to mobile users.

The cellular concept increases **system capacity, coverage, and frequency reuse**.

5.2 Cellular Concept

- The service area is divided into **hexagonal cells**
- Each cell has a base station
- Mobile users communicate with the nearest base station
- Adjacent cells use different frequencies to avoid interference

Advantages of Cellular Concept:

1. Efficient frequency reuse
2. Increased system capacity
3. Wide area coverage
4. Supports mobility

5.3 Cell Structure and Components

(a) Mobile Station (MS)

- User equipment (mobile phone)

- Contains SIM card

(b) Base Station (BS)

- Communicates with mobile stations
- Handles radio transmission

(c) Base Station Controller (BSC)

- Controls multiple base stations
- Manages handover and resource allocation

(d) Core Network

- Switching and call routing
- Billing and subscriber management

5.4 Frequency Reuse

Frequency reuse allows the **same frequency band to be used in different cells** separated by a sufficient distance. It is practice of using the same radio frequency channels in different cells of a cellular network to increase capacity.

Example: A 900 MHz frequency used in one cell can also be used in a distant cell with minimal interference.

Benefits:

- Increases system capacity
- Efficient use of spectrum

Frequency Reuse Factor:

- Ratio of total channels to channels per cell

5.5 Handover (Handoff)

Handover is the process of **transferring an active call or data session from one cell to another** as the user moves.

Types of Handover:

1. Hard handover – connection breaks before new one is made
2. Soft handover – connection maintained with multiple cells

5.6 Power Control

Power control adjusts the **transmission power of mobile devices** to:

- Reduce interference
- Save battery power
- Improve signal quality

5.7 Interference in Cellular Systems

Types of Interference:

- Co-channel interference
- Adjacent channel interference

Effects:

- Call drops
- Poor voice quality
- Reduced data rate

5.8 GSM Architecture (Overview)

Main Components:

- Mobile Station (MS)
- Base Station Subsystem (BSS)
- Network and Switching Subsystem (NSS)
- Operation and Support Subsystem (OSS)

Features of GSM:

- Digital communication
- Supports voice and SMS
- Uses TDMA

5.9 Introduction to 3G and 4G Systems

3G (UMTS):

- Higher data rate than GSM
- Supports video calling and mobile internet

4G (LTE):

- All-IP based system
- Very high data rates
- Supports multimedia services

5.10 Challenges of Mobile Computing Compared to Wired Computing

- **Limited bandwidth** – Wireless channels are slower and shared.
- **Battery constraints** – Mobile devices have finite power.
- **Security risks** – Wireless networks are more exposed to attacks.
- **Network variability** – Connectivity may drop due to movement, interference, or distance.
- **Hardware limitations** – Mobile devices have smaller memory and processing power.

MODULE 6

ANTENNAS AND DIVERSITY TECHNIQUES

6.1 Introduction

Antennas are devices used to **transmit and receive electromagnetic waves** in wireless communication systems.
Diversity techniques are used to **improve signal quality and reduce fading** by transmitting or receiving signals through multiple paths.

6.2 Basic Antenna Concepts

Definition of Antenna

An antenna is a **transducer** that converts electrical signals into electromagnetic waves (transmission) and electromagnetic waves into electrical signals (reception).

6.3 Types of Antennas

(a) Omnidirectional Antenna

- Radiates power equally in all directions
- Used in mobile phones and Wi-Fi routers

(b) Directional Antenna

- Radiates power in a specific direction
- Used in base stations and satellite communication

6.4 Antenna Parameters

(a) Gain

- Measure of how well an antenna directs energy
- Higher gain → longer transmission range

(b) Radiation Pattern

- Graphical representation of antenna radiation

(c) Bandwidth

- Range of frequencies an antenna can operate

(d) Polarization

- Orientation of electromagnetic wave
- Can be vertical, horizontal, or circular

6.5 Diversity Techniques

Diversity techniques improve communication reliability by **sending or receiving multiple copies of the signal** through different paths.

Types of Diversity

(a) Time Diversity

- Same signal transmitted at different times
- Reduces effect of deep fading

(b) Frequency Diversity

- Signal transmitted on different frequencies

(c) Space (Spatial) Diversity

- Uses multiple antennas separated in space

(d) Polarization Diversity

- Uses antennas with different polarizations

6.6 Multiple Input Multiple Output (MIMO)

MIMO uses **multiple transmitting and receiving antennas** to improve system performance.

Advantages of MIMO:

- Increased data rate
- Improved signal quality
- Better resistance to fading

Application:

- Wi-Fi
- 4G LTE
- 5G systems

6.7 Beamforming

Beamforming is a technique that **focuses the signal energy in a specific direction** instead of broadcasting in all directions.

Benefits:

- Increased signal strength
- Reduced interference
- Improved network capacity

MODULE 7

WIRELESS DATA AND NETWORK TECHNOLOGIES

7.1 Introduction

Wireless data and network technologies enable the **transmission of data without physical cables**. These technologies support internet access, short-range communication, and connectivity between devices in modern wireless systems.

7.2 Wireless Local Area Network (WLAN – Wi-Fi)

Wireless LAN provides **high-speed wireless connectivity within a limited area**, such as homes, schools, offices, and campuses.

Features of Wi-Fi:

- Based on IEEE 802.11 standards
- Operates in unlicensed frequency bands (2.4 GHz and 5 GHz)
- Supports high data rates

Types of WLAN

WLAN or more specifically the IEEE 802.11 is differentiated in terms of its operating modes. Types of WLAN are as follows:

- **Infrastructure**
- **Ad hoc:**
- **Bridge:**
- **Wireless distribution system:**

Components:

- Access Point (AP)
- Wireless devices (laptops, phones)

Products of WLAN

1. **Access Points (APs)** – Connect wireless devices to wired networks.
2. **Wireless NICs (Network Interface Cards)** – Hardware in devices enabling WLAN connectivity.
3. **Wireless Routers** – Combine AP and routing functionalities for home and small networks.
4. **Wireless Controllers** – Centralized devices that manage multiple APs in enterprise WLANs.

Advantages:

- Easy installation
- Mobility
- Cost-effective

7.3 Bluetooth Technology

Bluetooth is a **short-range wireless communication technology** used for connecting devices over short distances.

Features:

- Operates in 2.4 GHz band
- Low power consumption
- Short range (about 10 meters)

Applications:

- Headsets and speakers
- File transfer
- Wireless keyboards and mice

7.4 Worldwide Interoperability for Microwave Access (WiMAX)

WiMAX provides **wireless broadband access over long distances**.

Features:

- Based on IEEE 802.16
- Covers large geographical areas
- Supports high data rates

Applications:

- Internet access in rural areas
- Metropolitan networks

7.5 Internet of Things (IoT) Communication

IoT refers to **interconnection of everyday devices** through the internet.

Features:

- Uses sensors and smart devices
- Requires low power communication
- Supports real-time data transmission

Examples:

- Smart homes
- Smart agriculture
- Health monitoring systems

7.6 Wireless Personal Area Network (WPAN)

WPAN covers **short-range communication between personal devices**.

Examples:

- Bluetooth
- ZigBee

7.7 Comparison of Wireless Technologies

Technology	Range	Data Rate	Application
Wi-Fi	Medium	High	Internet access
Bluetooth	Short	Low	Device connection

Technology	Range	Data Rate	Application
WiMAX	Long	High	Broadband access
IoT	Varies	Low–Medium	Smart devices

7.8 Technical Principles of WLAN

1. Radio Frequency Transmission – WLAN uses RF signals (usually 2.4 GHz or 5 GHz bands) to transmit data between devices and access points.
2. Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) – Method to avoid data collisions in wireless medium. Devices sense the channel before transmitting.
3. SSID (Service Set Identifier) – Identifies and distinguishes different WLANs; ensures devices connect to the correct network.
4. Roaming Support – Allows devices to move between access points without dropping the connection.

7.9 Security in WLAN

- **WLAN security** protects data and devices from unauthorized access and attacks.
- Common measures:
    - **WEP/WPA/WPA2/WPA3** encryption standards.
    - **802.1X authentication** for enterprise networks.
    - **MAC address filtering** to control device access.
    - **Firewall and Intrusion Detection Systems (IDS)** for monitoring.

MODULE 8

MOBILITY MANAGEMENT AND QUALITY OF SERVICE (QoS)

8.1 Introduction

Mobility management ensures that **mobile users stay connected** to the network while moving from one cell or network to another.
Quality of Service (QoS) ensures that **wireless networks deliver data reliably** with acceptable speed, delay, and quality.

8.2 Mobility Management

Mobility management involves tracking the location of mobile users and managing **handoffs** between cells to maintain uninterrupted service.

Functions of Mobility Management:

1. Location management – keeps track of mobile user locations
2. Handoff management – switches active sessions from one cell to another

8.3 Location Management

Location management allows the network to **know where a mobile user is** for call setup and data delivery.

Components:

- **Home Location Register (HLR):** Database of subscriber info
- **Visitor Location Register (VLR):** Temporary info of users visiting a cell

Advantages:

- Efficient call routing
- Reduces dropped calls

8.4 Handoff (Handover) Management

Handoff is the process of **transferring an ongoing call or data session** from one base station to another.

Types of Handoff:

1. **Hard Handoff:** Connection breaks before a new one is established
2. **Soft Handoff:** Connection maintained with multiple cells during transition

Factors Affecting Handoff:

- Signal strength
- User speed
- Network load

8.5 Quality of Service (QoS)

QoS refers to **network performance in terms of reliability, delay, and throughput**. It ensures **users receive the expected service quality**.

QoS Parameters:

1. **Throughput:** Data transmitted per second
2. **Delay (Latency):** Time for data to reach destination
3. **Jitter:** Variation in delay
4. **Packet loss:** Percentage of lost data packets
5. **Reliability:** Consistency of service

8.6 Techniques to Improve QoS

1. **Traffic prioritization** – Give priority to voice or video over less critical data
2. **Resource allocation** – Reserve bandwidth for important services
3. **Error correction** – Reduce packet loss
4. **Load balancing** – Distribute traffic evenly across the network

8.7 Applications of Mobility Management and QoS

- Mobile telephony (voice calls)

- Mobile internet and video streaming
- Real-time applications (VoIP, online gaming)
- IoT devices requiring reliable connectivity

MODULE 9

WIRELESS SECURITY

9.1 Introduction

Wireless networks are **more vulnerable to attacks** compared to wired networks because signals travel through the air.
Wireless security ensures that **data, voice, and other communications are protected** from unauthorized access, interception, and tampering.

9.2 Security Challenges in Wireless Communication

1. **Eavesdropping:** Unauthorized listening to transmitted data
2. **Data modification:** Alteration of transmitted information
3. **Impersonation:** Fake devices pretending to be legitimate users
4. **Denial of Service (DoS):** Overloading network to disrupt service
5. **Man-in-the-middle attacks:** Intercepting and possibly altering communication
6. **Rogue Access Points** – Unauthorized APs set up to steal user information.

9.3 Authentication Techniques

Authentication ensures that **only authorized users can access the network**.

Methods:

1. **Password-based authentication** – Simple user verification
2. **SIM-based authentication** – Used in mobile networks (GSM, 3G, 4G)
3. **Two-factor authentication (2FA)** – Combines password with OTP or token

9.4 Encryption Techniques

Encryption converts data into a **coded form** that can only be read by authorized receivers.

Types of Encryption:

- **Symmetric key encryption** – Same key for encryption and decryption
- **Asymmetric key encryption** – Uses public and private keys
- **WEP, WPA, WPA2** – Used in Wi-Fi security

Importance:

- Prevents eavesdropping
- Protects sensitive data
- Ensures confidentiality

9.5 SIM-Based Security

- SIM (Subscriber Identity Module) card **authenticates mobile users**
- Contains unique identifiers (IMSI) and encryption keys
- Protects against **SIM cloning** and unauthorized network access

9.6 Virtual Private Networks (VPNs) in Wireless Communication

VPN creates a **secure tunnel over a public network** for data transmission.

Features:

- Data encryption
- Authentication
- Ensures confidentiality and integrity

Applications:

- Remote access to corporate networks
- Secure mobile internet access

9.7 Best Practices in Wireless Security

1. Use strong passwords and change them regularly
2. Enable encryption (WPA2/WPA3) on Wi-Fi networks
3. Regularly update software and firmware
4. Use VPNs when accessing public networks
5. Limit access to trusted devices

9.8 Basic Components of Wireless Security

1. **Authentication** – Verifying that a user or device is allowed to access the network (e.g., WPA2, 802.1X).
2. **Encryption** – Protecting transmitted data from interception (e.g., WPA2 uses AES encryption).
3. **Access Control** – Restricting network access based on roles, devices, or locations.
4. **Monitoring and Auditing** – Tracking network activity to detect unusual behaviour and potential threats.

9.9 Vulnerabilities on Wireless Security

1. **Open Networks** – No encryption, allowing free access.
2. **Weak Encryption** – Using outdated WEP or weak passwords.
3. **Rogue Access Points** – Unauthorized APs posing as legitimate ones.
4. **Man-in-the-Middle (MITM)** – Attacker intercepts and alters data traffic.
5. **Denial of Service** – Signal jamming or flooding network to block legitimate users.

MODULE 10

EMERGING WIRELESS TECHNOLOGIES

10.1 Introduction

Emerging wireless technologies refer to **new and advanced wireless systems** that improve data speed, capacity, reliability, and enable new applications such as IoT, smart cities, and autonomous devices.
These technologies include **5G, millimeter-wave communication, massive MIMO, cognitive radio, and software-defined radio (SDR)**.

10.2 Fifth Generation (5G) Wireless Technology

5G is the **latest generation of cellular networks**, designed to provide **very high data rates, low latency, and massive connectivity**.

Features:

- Peak data rates up to 20 Gbps
- Ultra-low latency (<1 ms)
- High device density (IoT support)
- Enhanced mobile broadband and ultra-reliable communication

Applications:

- Autonomous vehicles
- Smart cities
- Remote surgery and healthcare
- Virtual and augmented reality

10.3 Millimeter-Wave (mmWave) Communication

- Uses **high-frequency spectrum (30–300 GHz)**
- Provides **high data rates** due to wide bandwidth
- Limited coverage due to **high path loss and obstacles**

Applications:

- 5G small cells
- High-speed wireless links
- Indoor ultra-fast networks

10.4 Massive MIMO (Multiple Input Multiple Output)

- Uses **hundreds of antennas** at base stations
- Allows **simultaneous communication with multiple users**
- Improves **capacity, data rate, and energy efficiency**

Advantages:

- Increased spectral efficiency
- Reduced interference
- Supports dense networks

10.5 Cognitive Radio

Cognitive radio is an **intelligent wireless system** that can:

- Sense the spectrum
- Detect unused frequencies (spectrum holes)
- Adapt transmission to avoid interference

Advantages:

- Efficient spectrum utilization
- Reduces congestion
- Supports dynamic spectrum access

10.6 Software-Defined Radio (SDR)

SDR is a **flexible radio system** where software performs functions traditionally done by hardware.

Features:

- Modulation and demodulation implemented in software
- Supports multiple wireless standards
- Easily upgradable

Applications:

- Military communications
- Wireless research and prototyping
- Multi-standard mobile devices

10.7 Introduction to Sixth Generation (6G) Concepts

- 6G is **still under research**, expected around 2030
- Expected features:
    - Terabit-per-second data rates
    - Extremely low latency
    - Integration with AI and satellite networks
- Applications: holographic communication, autonomous networks, extreme IoT

How a Bridge Differs from a Repeater

Feature	Bridge	Repeater
Function	Connects two LAN segments and filters traffic	Amplifies signals to extend coverage
Network Layer	Operates at Data Link Layer (Layer 2)	Operates at Physical Layer (Layer 1)
Traffic Filtering	Yes; forwards only required traffic	No; retransmits all signals
Use Case	Connects different network segments or buildings	Extends signal range without filtering

ADVANCED WIRELESS COMMUNICATIONS – OUTLINE QUESTIONS

MODULE 1 – Introduction to Wireless Communications

1. Define wireless communication.
2. List five advantages of wireless communication.
3. State three limitations of wireless communication.
4. Explain the evolution of wireless communication from 1G to 5G.
5. List the main components of a wireless communication system.
6. Name two regulatory bodies responsible for spectrum allocation.

MODULE 2 – Wireless Propagation & Channel Characteristics

1. Explain the free-space propagation model.
2. Define path loss and state its causes.
3. What is shadowing in wireless communication?
4. Explain multipath propagation.
5. Differentiate between fast fading and slow fading.
6. Define Doppler effect in wireless systems.
7. List three sources of noise in wireless communication.

MODULE 3 – Digital Modulation Techniques

1. Define digital modulation.
2. Explain Amplitude Shift Keying (ASK).
3. Explain Frequency Shift Keying (FSK).
4. Explain Phase Shift Keying (PSK) and Quadrature PSK (QPSK).
5. What is Quadrature Amplitude Modulation (QAM)?
6. Define Bit Error Rate (BER).
7. List two advantages of OFDM.

MODULE 4 – Multiple Access Techniques

1. Define multiple access technique.
2. Explain Frequency Division Multiple Access (FDMA).
3. Explain Time Division Multiple Access (TDMA).
4. Explain Code Division Multiple Access (CDMA).
5. Explain Orthogonal Frequency Division Multiple Access (OFDMA).
6. Compare FDMA, TDMA, CDMA, and OFDMA.

MODULE 5 – Cellular Communication Systems

1. Define cellular communication system.

2. Explain the cellular concept.
3. What is frequency reuse?
4. Define handover and state its types.
5. Explain power control in cellular systems.
6. List the main components of GSM architecture.
7. State two differences between 3G and 4G systems.

MODULE 6 – Antennas & Diversity Techniques

1. Define antenna.
2. Differentiate between omnidirectional and directional antennas.
3. List four antenna parameters.
4. Define diversity techniques.
5. List types of diversity (time, frequency, spatial, polarization).
6. Explain MIMO (Multiple Input Multiple Output).
7. What is beamforming and its advantage?

MODULE 7 – Wireless Data & Network Technologies

1. Define wireless data communication.
2. Explain WLAN (Wi-Fi) technology.
3. State two applications of Bluetooth.
4. Define WiMAX and its uses.
5. Define Internet of Things (IoT) and give examples.
6. Differentiate Wi-Fi and Bluetooth.
7. What is a WPAN?

MODULE 8 – Mobility Management & Quality of Service (QoS)

1. Define mobility management.
2. Explain the function of Home Location Register (HLR) and Visitor Location Register (VLR).
3. Define handover and differentiate between hard and soft handoff.
4. Define Quality of Service (QoS).
5. List five QoS parameters.
6. Mention two techniques to improve QoS.
7. Give two applications of mobility management.

MODULE 9 – Wireless Security

1. List four security challenges in wireless communication.

2. Explain authentication techniques in wireless networks.
3. Differentiate symmetric and asymmetric encryption.
4. How does SIM-based security protect mobile users?
5. What is a VPN and why is it used?
6. Mention three best practices to improve wireless security.

MODULE 10 – Emerging Wireless Technologies

1. List three features of 5G wireless technology.
2. What is millimeter-wave (mmWave) communication and its main limitation?
3. Explain massive MIMO and its advantage.
4. Define cognitive radio and state one benefit.
5. What is software-defined radio (SDR)?
6. Mention two expected features of 6G wireless networks.
7. Give two applications of 5G technology.